Agenda

* Recap
* Sensitivity and specificity
* ROC
* AUC
* Precision Recall curve
* Imbalance Data
* SMOTE

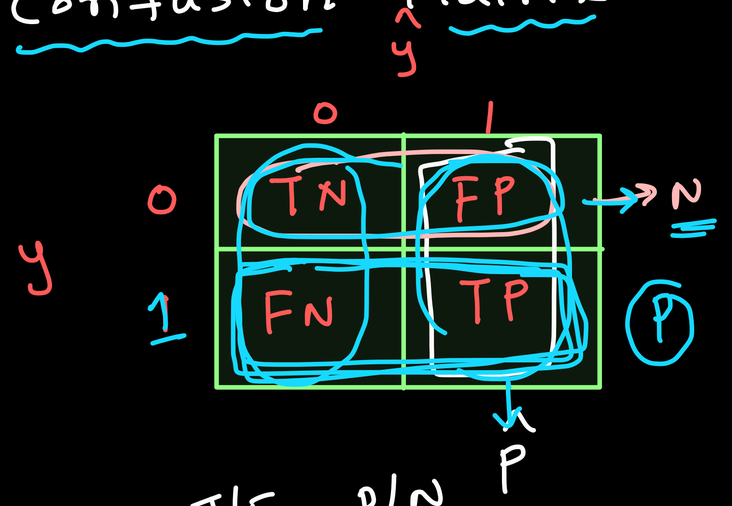
Assignments

7780568417

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Recap

Confusion Matrix



T/F — P/N → prediction

Precision

$$\text{precision} = \frac{TP}{TP+FP}$$

TP↑ FP↓

Recall

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{TP}{P}$$

TP↑ FN↓

F1-Score

$$F1 = \frac{2PR}{P+R}$$

Sensitivity.

TPR / TNR

✓✓ TPR → True positive Rate $= \frac{TP}{\text{all positive}} = \frac{TP}{TP+FN} =$ Recall

↳ Goal : TP ↑ FN ↓

✓✓ TNR → True Negative Rate $= \frac{TN}{\text{all negative}} = \frac{TN}{TN+FP}$

TPR → Sensitivity

TNR → Specificity ⇒ TN ↑ FP ↓

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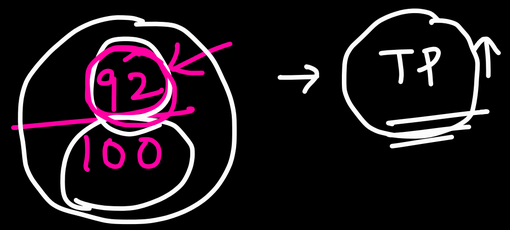
Quiz #1

What to say when screening test identifies 92 Cancer patients out of 100?

2 users have participated

✓ A	test has high sensitivity	100%
B	test has low sensitivity	0%
C	test has no sensitivity	0%
D	cannot be determined	0%

End Quiz Now



1 positive → cancer

0 Ne → Not-cancer

FN ↓

FPR = FP / all negative = FP / (FP + TN)

Incorrectly

all healthy patients how many patients you have detected inccorrectly

FNR = FN / all positive = FN / (FN + TP)

ROC [Receiver operating characteristic]

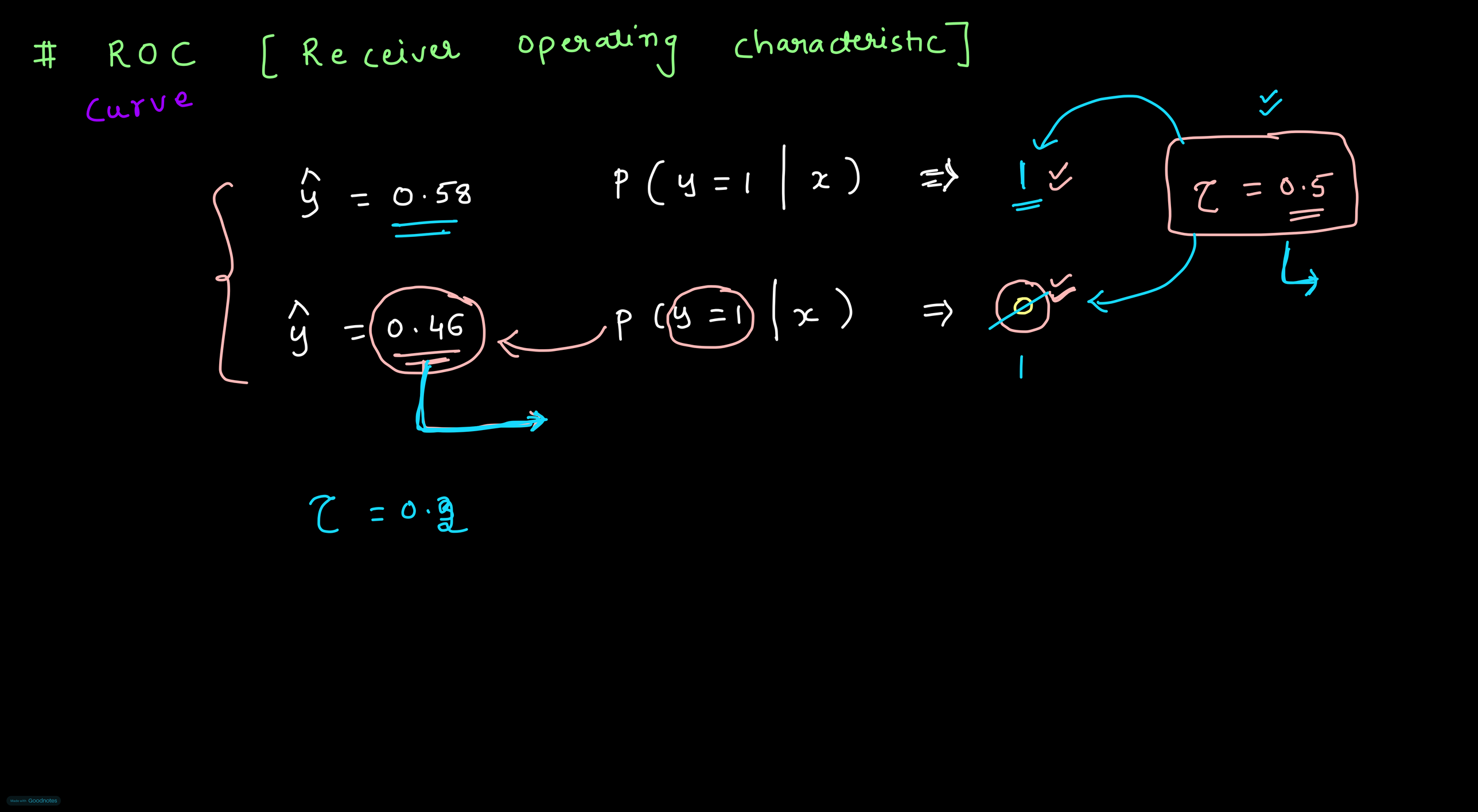
Curve

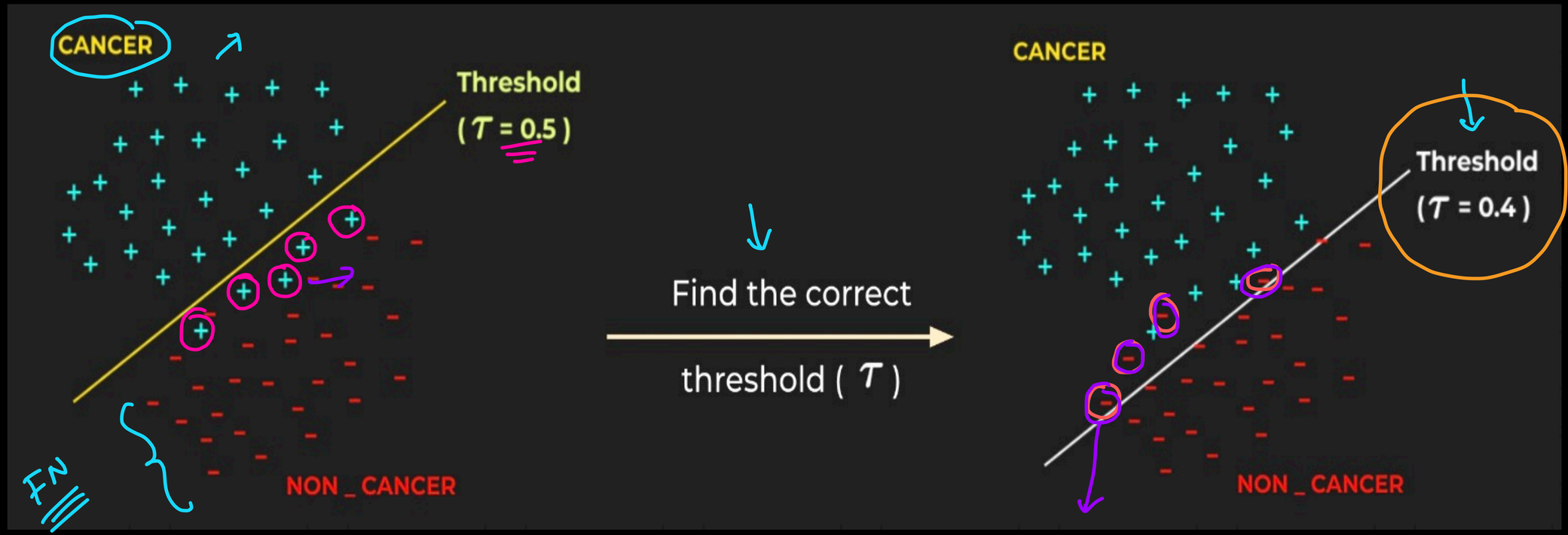
$\hat{y} = 0.58$ $\quad P(y=1 \mid x) \Rightarrow 1$

$\hat{y} = 0.46$ $\quad P(y=1 \mid x) \Rightarrow 0$

$\tau = 0.5$

$\tau = 0.3$





FP

(1)

Minimize FN

(2)

Dataset

Step1

X	Y	P
$x^{(1)}$	1	0.65
$x^{(2)}$	1	0.94
$x^{(3)}$	0	0.30
$x^{(4)}$	1	0.92
$x^{(5)}$	0	0.70
$x^{(6)}$	0	0.20

Sort the data → descending order prob

X	Y	P
$x^{(2)}$	1	0.94
$x^{(4)}$	1	0.92
$x^{(5)}$	0	0.70
$x^{(1)}$	1	0.65
$x^{(3)}$	0	0.30
$x^{(6)}$	0	0.20

Step2

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Step2 ⇒ unique probabilities, consider them as a threshold and find the prediction.

X	Y	P	$\hat{y}_\tau = 0.94$	$\hat{y}_\tau = 0.92$	$\hat{y}_\tau = 0.20$
$x^{(2)}$	1	0.94	1 ✓	1 ✓	1 ✓
$x^{(4)}$	1	0.92	0	1 ✓	1 ✓
$x^{(5)}$	0	0.70	0	0	1 ✓
$x^{(1)}$	1	0.65	0	0	1 ✓
$x^{(3)}$	0	0.30	0	0	1 ✓
$x^{(6)}$	0	0.20	0	0	1 ✓

→ Roc curve

✓ $TPR^{(1)}$ $TPR^{(2)}$ - - - $TPR^{(n)}$

✓ $FPR^{(1)}$ $FPR^{(2)}$ $FPR^{(n)}$

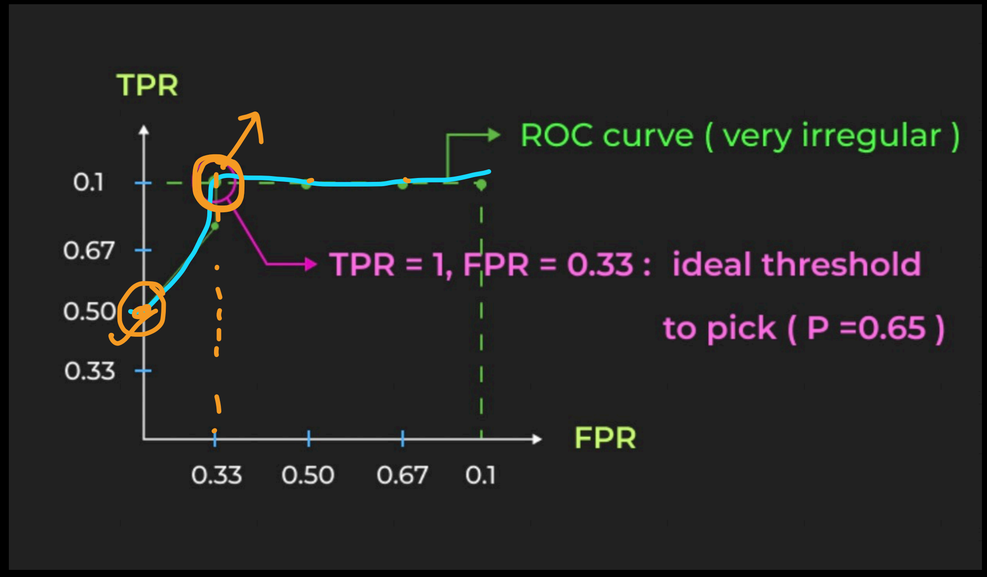
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P	TPR	FPR
0.94	0.33	0
0.92	0.50	0
0.70	0.67	0.33
0.65	1	0.33
0.30	1	0.67
0.20	1	1

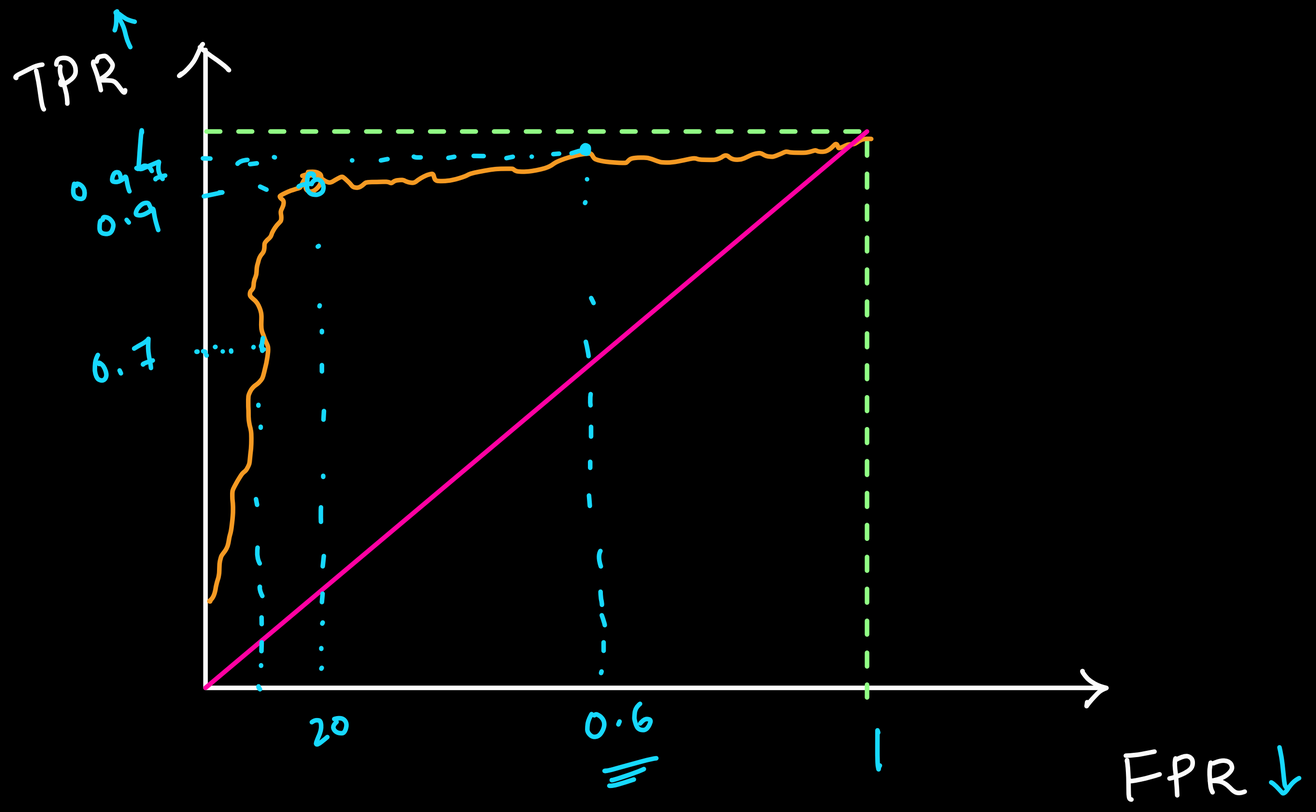
TPR → ↑

FPR → ↓

Roc - Curve



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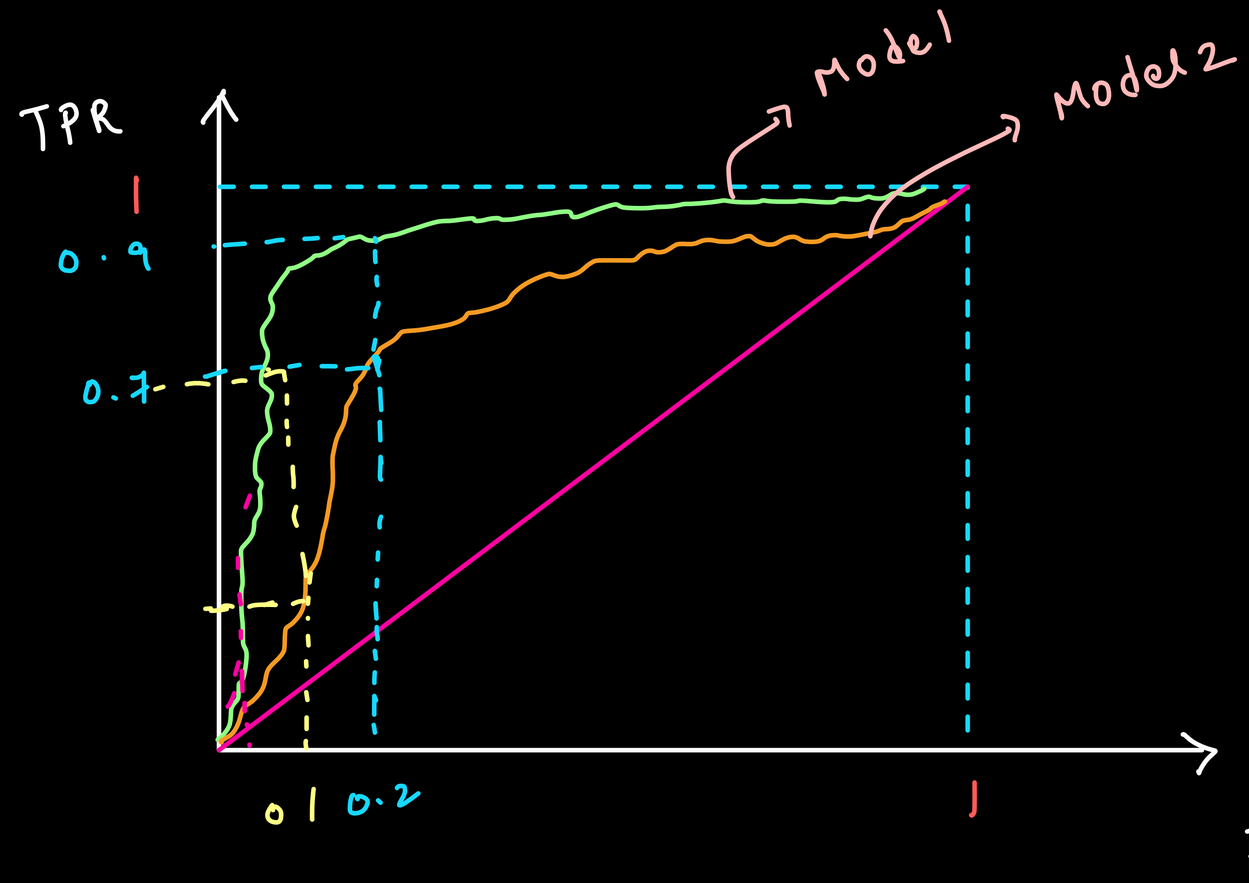


Random Model

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→ Model 1

→ Model 2



How many points are typically used to plot an ROC curve?

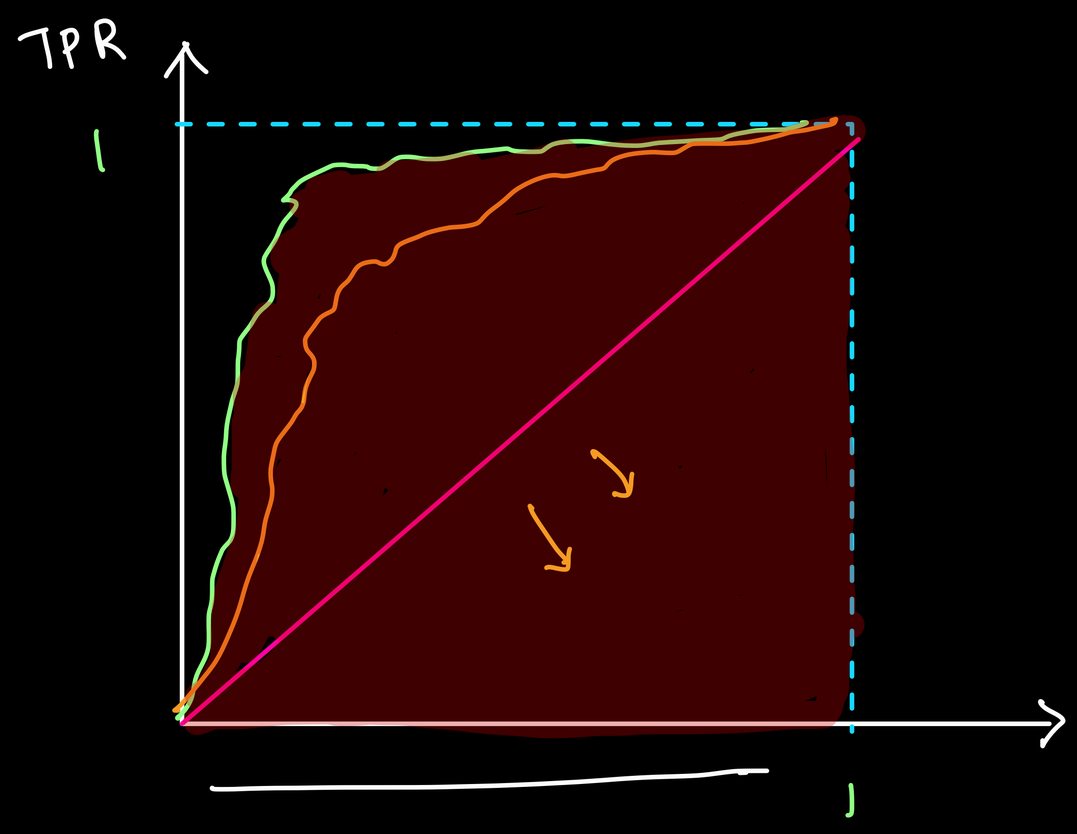
1 user has participated

A	2 points (0,0) and (1,1)	0%
B	3 points representing the thresholds 0.25, 0.5, and 0.75	0%
C	10 points equally spaced between 0 and 1	0%
D	Depends on the number of unique threshold values	100%

End Quiz Now

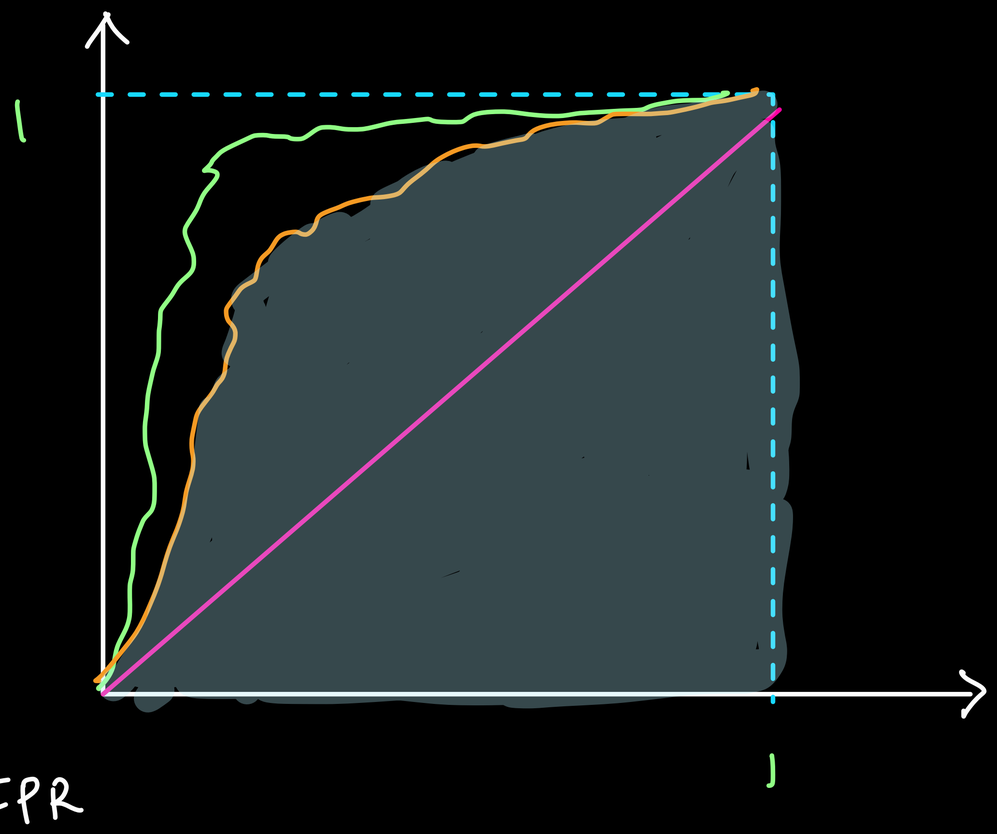
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AU – ROC / AUC



Model 1

90



Model 2

80

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$$= \frac{1}{2} * b * h$$

$$\text{AUC (Random)} = \frac{1}{2} * 1 * 1 = 0.5$$

10:13 → 10:18

Which of the following metrics can be directly derived from the ROC curve?

1 user has participated

A	Accuracy	0%
B	Precision	0%
C	Recall	100%
D	Area Under the Curve (AUC)	0%

End Quiz Now

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Issue with AUC

1. Binary classes → not for multiclass
2. Imbalanced data → AUC is not good metric
3. ordering of probability matters & not actual probabilities

M1

y	P_{m1}	$\hat{y}_\tau = 0.95$	$\hat{y}_\tau = 0.92$
1	0.95	1	1
1	0.92	0	1
0	0.80	0	0
1	0.76	0	0
1	0.71	0	0

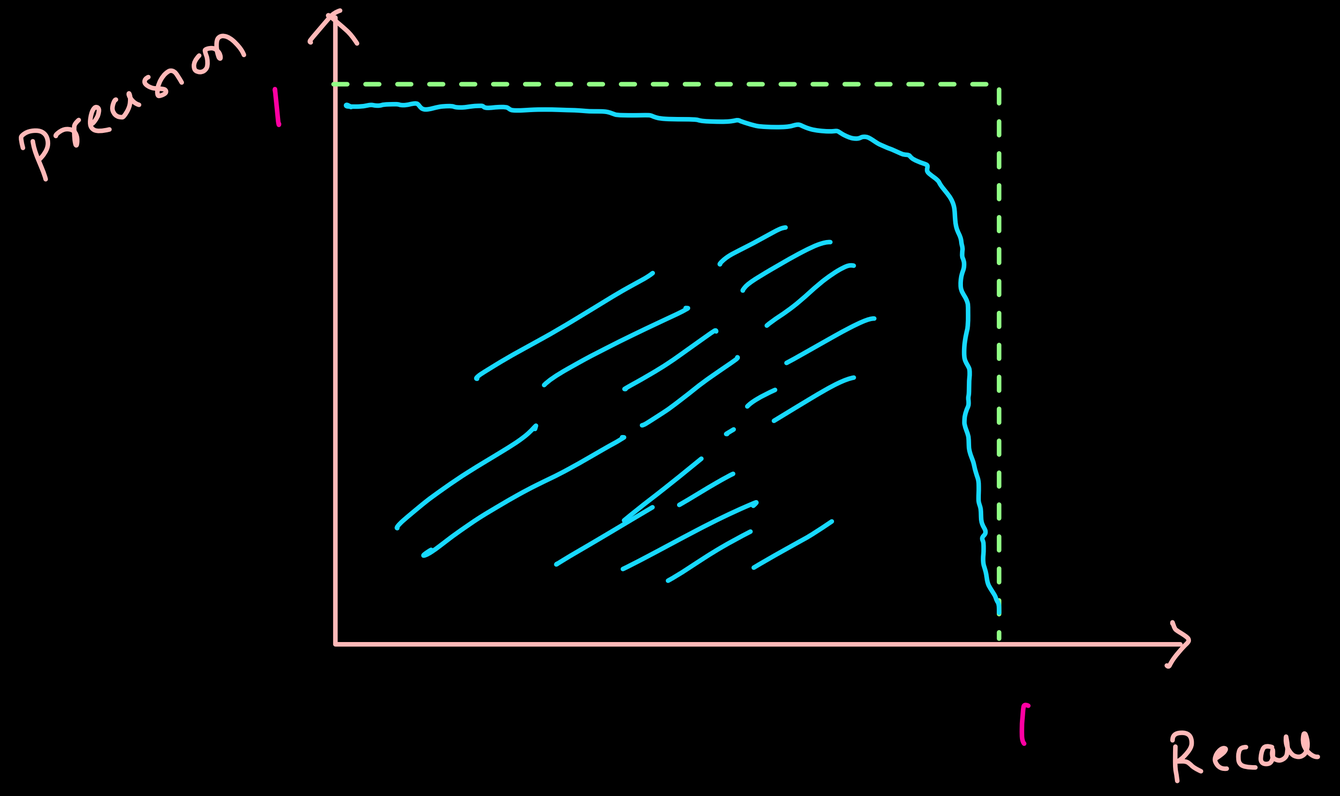
M2

y	P_{m2}	$\hat{y}_\tau = 0.2$	$\hat{y}_\tau = 0.1$
1	0.2	1	1
1	0.1	0	1
0	0.08	0	0
1	0.06	0	0
1	0.02	0	0

FPR =
TPR =

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Precision Recall Curve



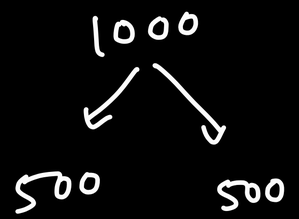
F1-score

Precision

Recall

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Imbalanced Data



50% — 50% ⟶ balanced data

60 — 40 ⟶ Almost balanced

70 — 30 ⟶ Imbalanced data

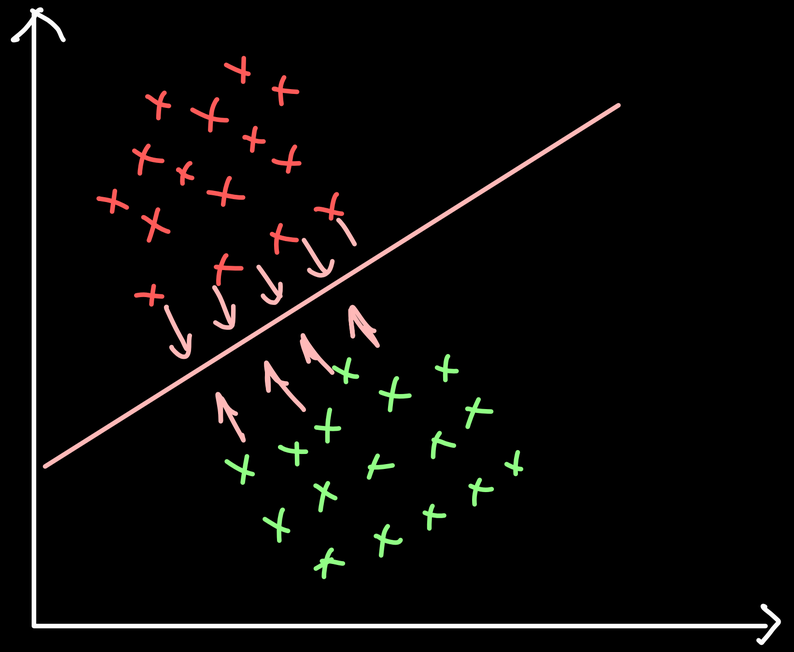
80 — 20 ⟶ High Imbalanced

95 — 5 ⟶ Extreamly high "

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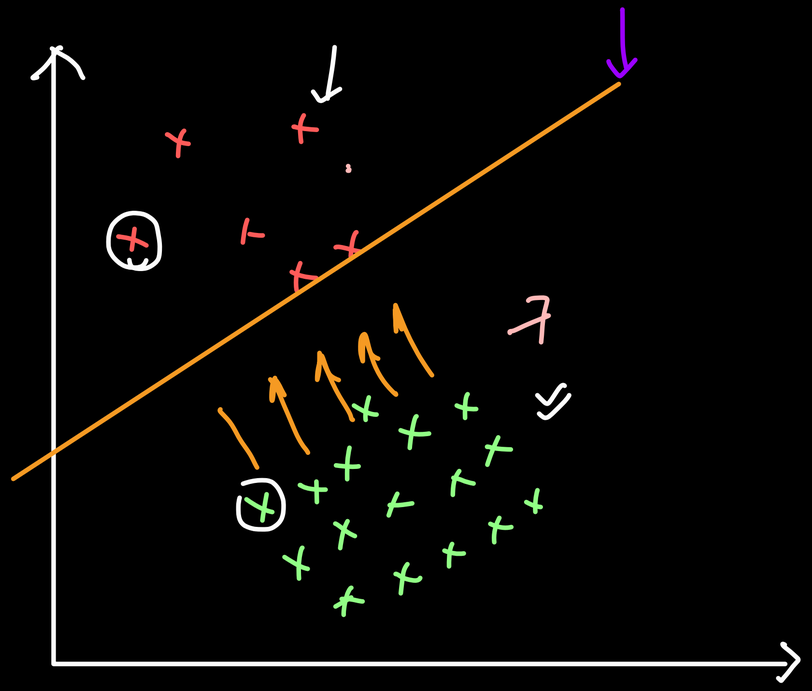
Case 1

Balanced



Case 2

Imbalanced



1000 data points

850

N.S

150

S

weight $= \frac{850}{150} = \simeq 5.6$

$w_R = 5.6$

$w_g = 1$

$$\text{loss} = \sum_{i=1}^{n} \text{log loss}_i * w_i^{5.6} + \sum_{j=1}^{d} w_j^2$$

weight

class_weight = { 0 : 1 , 1 : 5 6 } class weights

= " balanced "

Why are class weights important in imbalanced datasets?

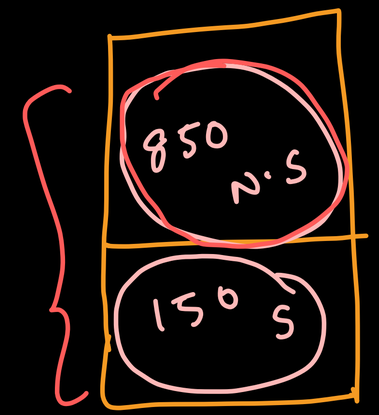
1 user has participated

A	They improve the convergence speed of the model	0%
B	They reduce overfitting in the model	0%
C	They help give more importance to the minority class	0%
D	They enhance the model's ability to handle missing values	100%

End Quiz Now

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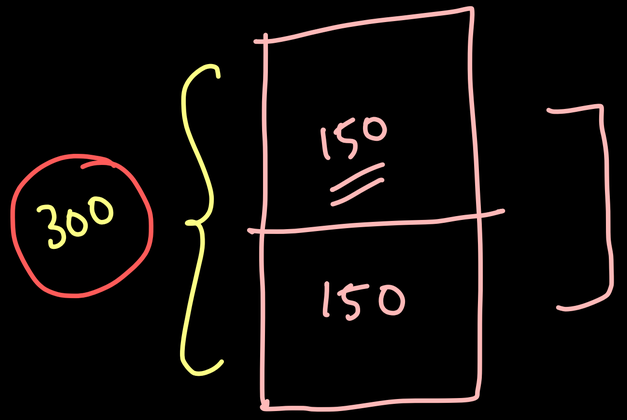
Undersampling



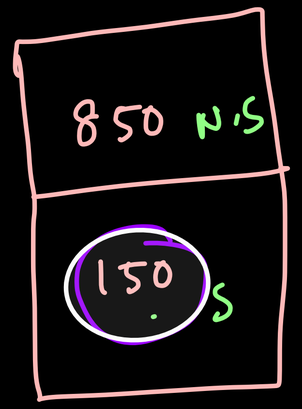
1000

Drawback:

loss of data

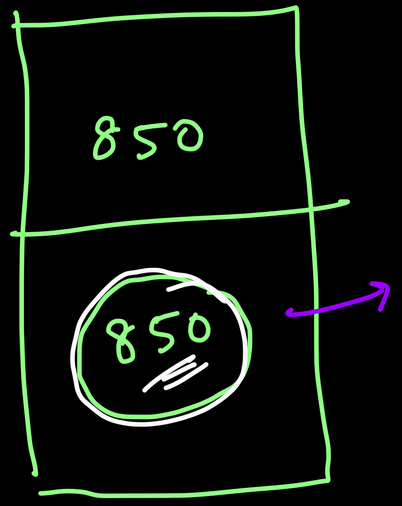


oversampling



Drawback:

lot of duplicates | Data quality



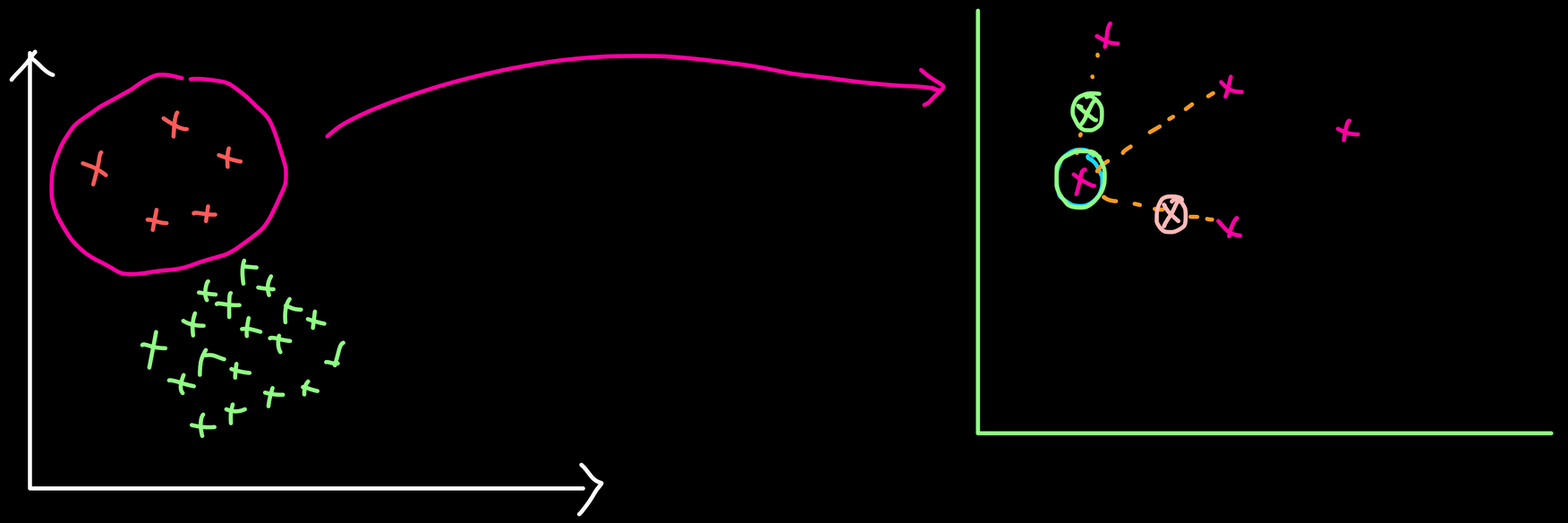
bootstrapping

sampling with replacement

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SMOTE

[Synthetically Minority Oversampling Technique Estimation]



1. Select any random point from minority class
2. Find K = 3 Nearest points
3. generate some random no b/w (0, 1)

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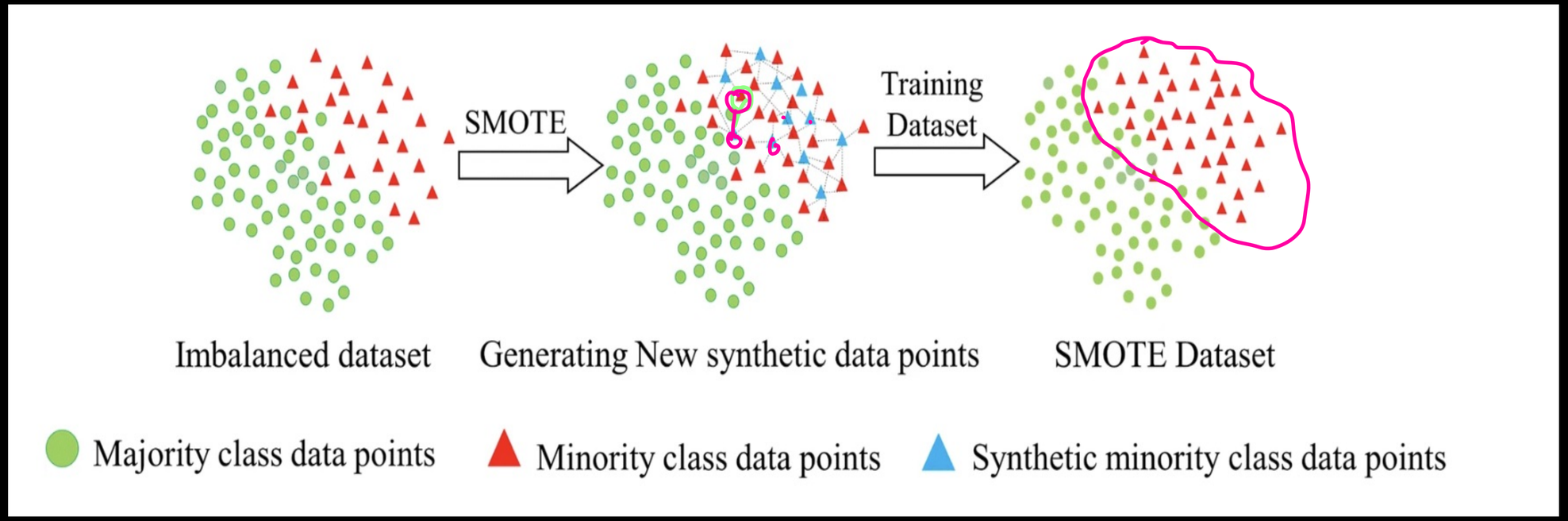
4. $x_{new1} = x + \alpha \cdot d(x, x_{N1})$

$\alpha \rightarrow 0.8 \times$

$d(x, x_{N1}) \rightarrow 5$

$x + \alpha \cdot d(x, x_{N2})$

$\alpha \rightarrow x + 0.2 * 7$



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